

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Data Visualization with Power BI and Tableau

Practical – I

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Class: TYIT	Batch: 1
Date of Assignment: 24-07-2026	Date/Time of Submission: 24-07-2026

Part A - Theory:

a) Connect to a CSV Dataset and Explore the Tableau Interface

Tableau allows users to connect to CSV files for data analysis. After loading the dataset, users can explore the Data pane, Sheets, Rows, Columns, Marks card, and Toolbar to create visualizations efficiently.

b) Identify and Differentiate Measures and Dimensions

Dimensions are qualitative fields (e.g., Name, Category, Region) used for grouping and categorizing data. Measures are quantitative fields (e.g., Sales, Profit, Quantity) used for calculations and analysis.

c) Convert Fields Between Discrete and Continuous

In Tableau, fields can be displayed as **Discrete (blue)** or **Continuous (green)**. Discrete fields create headers and separate categories, while continuous fields create axes for numerical analysis, changing the visualization style.

d) Create and Customize a Basic Bar Chart

A bar chart is used to compare values across different categories. Users can customize the chart by changing colors, sorting data, adding labels, adjusting sizes, and formatting axes for better readability.

e) Build a Line Chart for Time-Series Analysis

A line chart displays trends over time by connecting data points with lines. It is useful for analyzing changes, patterns, and growth across dates or time periods.

f) Create a Geographic Map Visualization

Tableau automatically recognizes geographic fields such as Country, State, or City and plots them on a map. Maps help visualize regional patterns, distributions, and location-based insights.

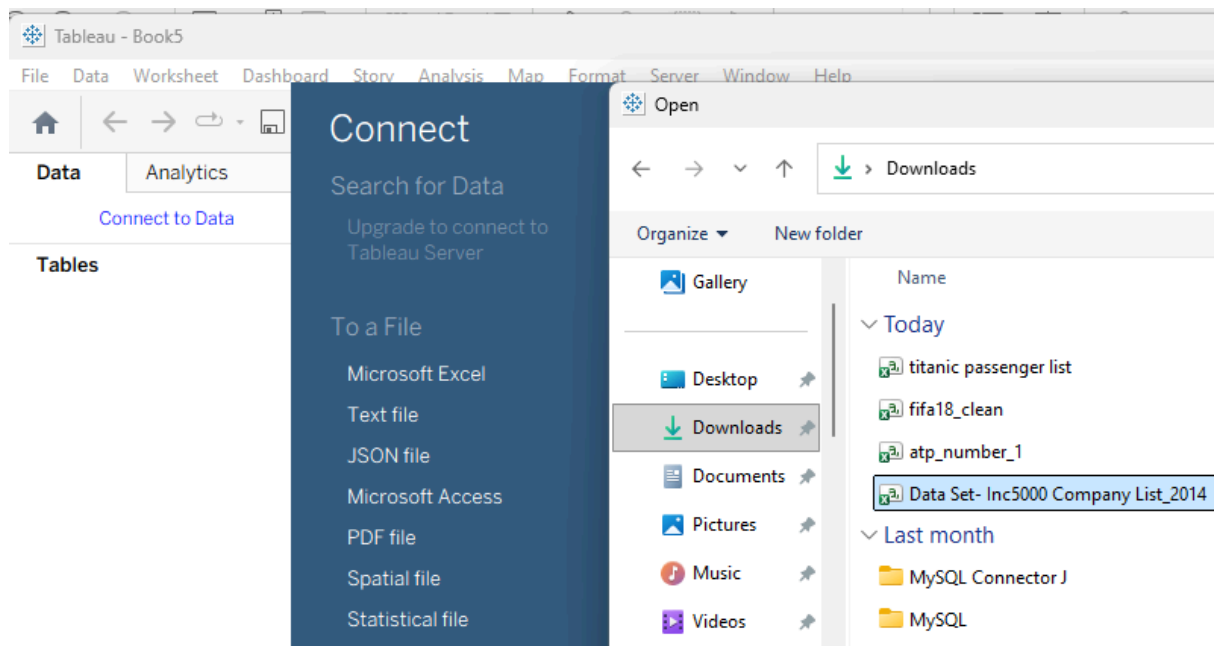
g) Combine Multiple Charts into a Simple Interactive Dashboard

A dashboard combines multiple worksheets into a single interactive view. Filters, actions, and highlights can be added so users can explore and compare data across different visualizations in one place.

Part B – Program

Getting Started with Tableau and Building Basic Visualizations

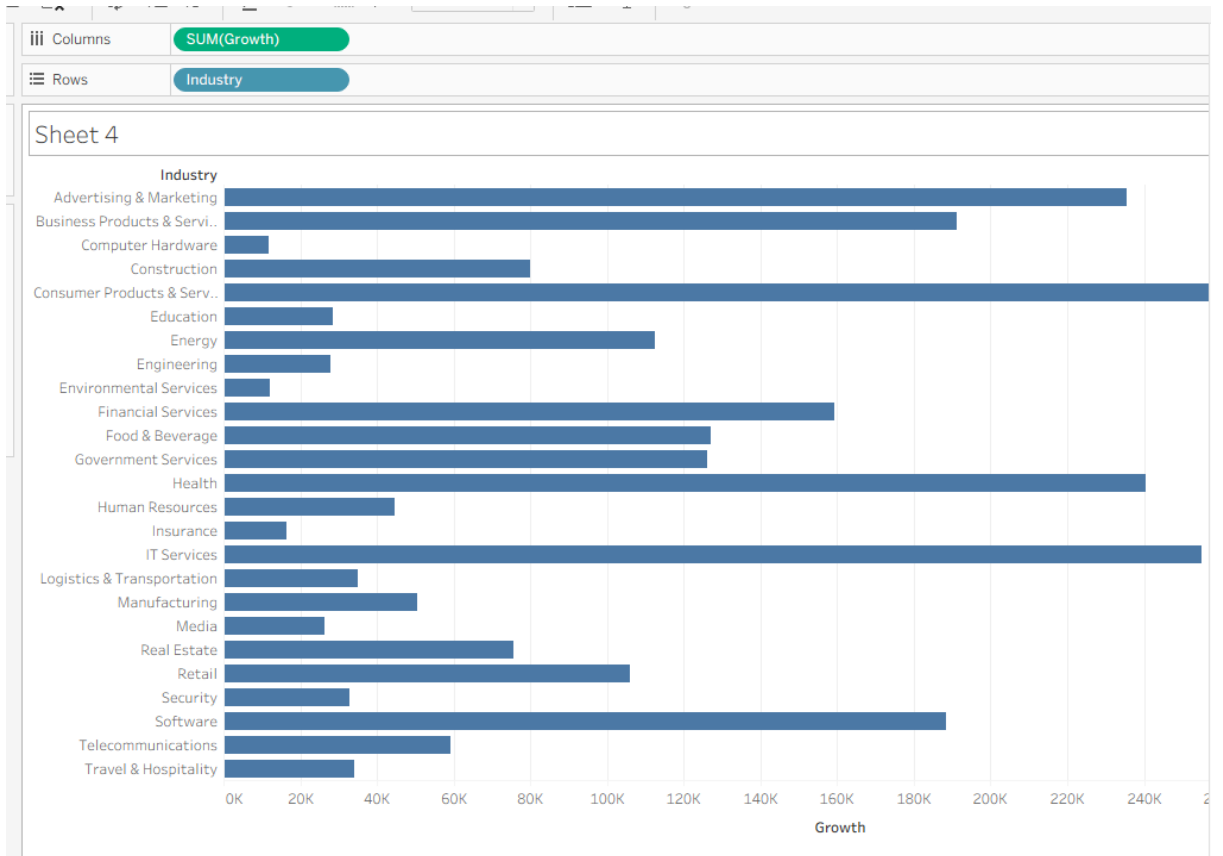
- a) Connect to a CSV dataset and explore the Tableau interface



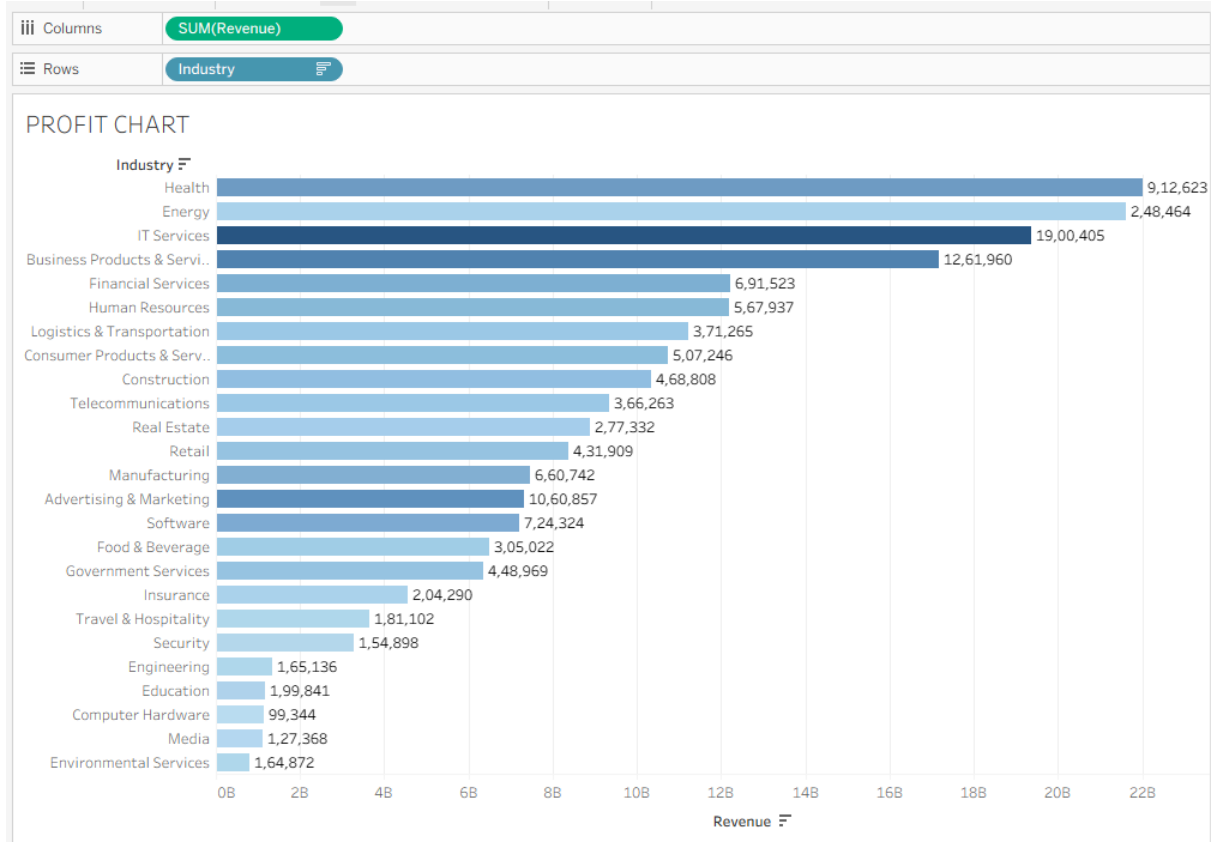
b) Identify and differentiate Measures and Dimensions

Search		Measure Names	
Tables			
Abc	_Input	#	Growth
#	_Num	#	Rank
Abc	_pageUrl	#	Revenue
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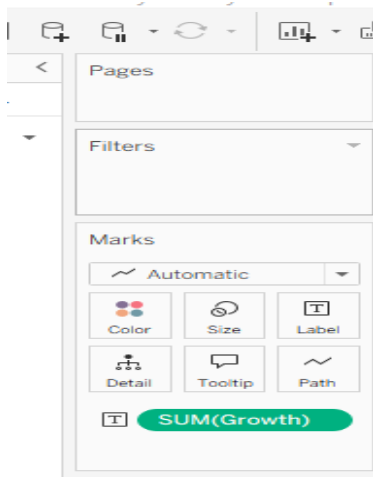
c) Convert fields between discrete and continuous and observe changes



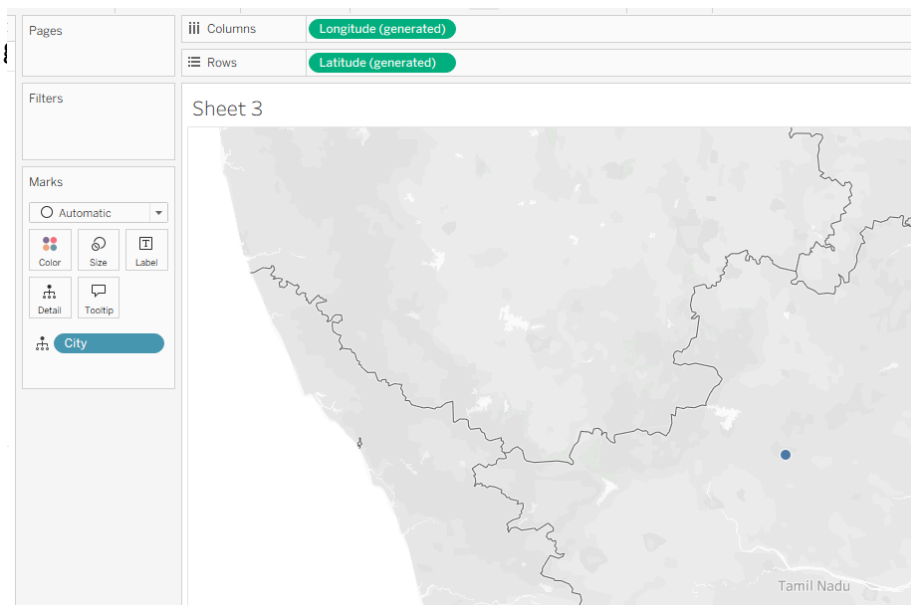
d) Create and customize a basic bar chart



e) Build a line chart for time-series analysis



f) Create a geographic map visualization



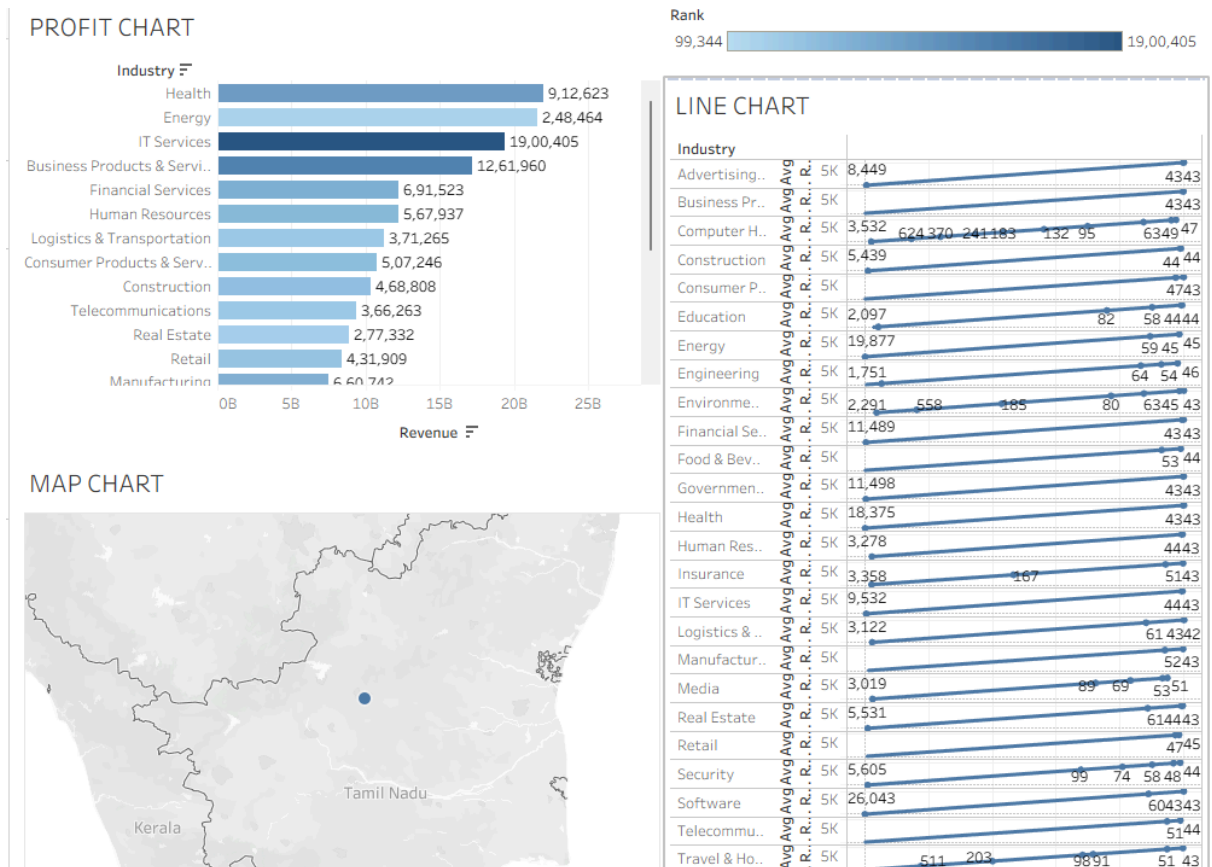


Tableau Dashboard Report

Title: Industry Revenue Analysis Dashboard

Objective

The objective of this dashboard is to analyze revenue across different industries using various Tableau visualizations. The dashboard includes a bar chart, line chart, and map chart to identify top-performing industries, observe trends, and understand the geographical distribution of the data.

Dashboard Components

1. Profit (Revenue) Bar Chart

Compares the revenue generated by different industries.

IT Services has the highest revenue (approximately 19 million).

Business Products & Services and Health are also among the top-performing industries.

Energy records the lowest revenue among the industries shown.

2. Line Chart

Displays the trend of the selected measure across industries.

Most industries show an upward trend, indicating steady growth or improvement in the selected metric.

The chart helps compare performance patterns between industries.

3. Map Chart

Shows the geographical location of the data.

The visualization indicates that the business activity is concentrated in a specific region, allowing users to identify the location associated with the dataset.

Observations

1. IT Services is the highest revenue-generating industry, contributing approximately 19 million, making it the leading sector in the dataset.
2. Business Products & Services, Health, and Financial Services also generate high revenue, indicating strong market performance.
3. Energy has the lowest revenue, suggesting comparatively weaker performance than other industries.
4. The line chart shows a generally increasing trend across industries, indicating positive growth in the selected measure.
5. The map chart provides geographical insight, helping users identify where the recorded business data is located.

Conclusion

The Tableau dashboard effectively presents industry-wise revenue analysis through multiple visualizations. It highlights the strongest and weakest performing industries, reveals overall trends, and provides geographical context. These insights can help organizations identify high-performing sectors, monitor business performance, and support informed decision-making.

Part C – Curiosity Question

Curiosity Question

Question:

Using a dataset of your choice (CSV format), create an interactive Tableau dashboard that effectively presents key insights from the data.

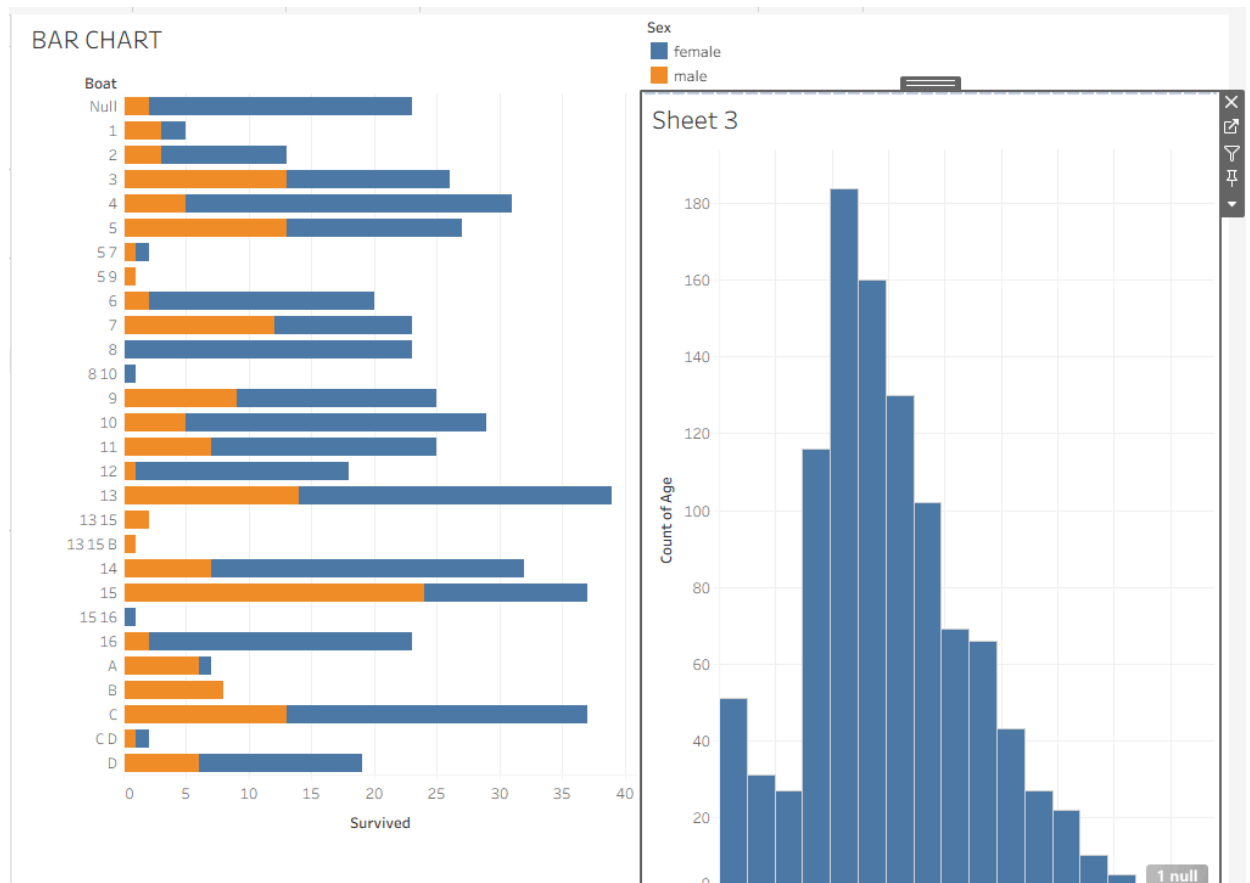
Requirements:

1. Import your own dataset into Tableau.
2. Identify appropriate **Dimensions** and **Measures**.
3. Create at least **three different visualizations** (e.g., bar chart, line chart, map, pie chart, scatter plot, etc.).

4. Add at least **one filter** or interactive element to the dashboard.
5. Arrange the charts into a **single interactive dashboard** with a clear title.
6. Write **3–5 observations** explaining the insights gained from your dashboard.

Submission:

- Tableau Workbook (.twb or .twbx)
- Dataset (.csv)
- Screenshot of the completed dashboard



- A short report containing the observations.

Tableau Dashboard Report

Title: Titanic Passenger Survival Analysis

Objective

The objective of this project is to analyze the Titanic dataset using Tableau and create an interactive dashboard to understand passenger survival patterns based on lifeboats, gender, and age distribution.

Dashboard Components

Bar Chart: Survival Count by Lifeboat and Gender

Histogram: Passenger Age Distribution

Interactive Filter: Gender (Sex)

Observations

1. **Female passengers had a higher survival count than male passengers** in most lifeboats, indicating that women were rescued more frequently.
2. **Lifeboats 13 and 15 recorded the highest number of survivors**, while several other lifeboats carried only a few passengers.
3. **Most passengers were between 20 and 35 years old**, making this the largest age group on the Titanic.
4. **The number of passengers decreases as age increases**, with very few passengers above 60 years of age. The histogram also indicates one missing age value.
5. **The interactive Gender filter** allows users to compare survival patterns between male and female passengers, making the dashboard easier to explore and analyze.